Feature extraction is a crucial step in many machine learning and data processing tasks. It involves transforming raw data into a set of features that can be used for model training and evaluation. There are various techniques for feature extraction, depending on the type of data you're working with. Here's a rundown of some common methods:

Dimensionality Reduction Techniques

- **Principal Component Analysis (PCA)**:
  Reduces data dimensions by projecting onto principal components.
- **Linear Discriminant Analysis (LDA)**:
  Maximizes class separability by projecting onto discriminant axes.
- **t-Distributed Stochastic Neighbor Embedding (t-SNE)**:
  Reduces dimensions while preserving the local structure of data.
- **Autoencoders**: Neural networks that learn a compressed representation of the input data.